



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	ELECTRODYNAMICS II				
Type of Course	Core in Major				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	PHY4CJ203- Electrodynamics I				
Course Summary	The course emphasizes the development of mathematical techniques such as vector calculus and differential equations to solve complex problems in electromagnetism. Through theoretical discussions, problem-solving sections, and possibly laboratory experiments, students gain a deep understanding of electromagnetic phenomena and their applications in various fields such as optics, electronics, and telecommunications.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain the fundamental principles of electromagnetism and Maxwell's equations	Understanding	Conceptual Knowledge	Written exams, quizzes
CO2	Apply mathematical techniques such as vector calculus and differential equations to solve electromagnetic problems	Applying	Procedural Knowledge	Problem sets, simulations
CO3	Analyze the behavior of electromagnetic fields in various media and under different boundary conditions	Analyzing	Conceptual Knowledge	Homework assignments, exams
CO4	Derive and interpret the electromagnetic wave equation and its solutions	Understanding	Conceptual Knowledge	Class discussions, presentations
CO5	Predict and analyze the behavior of electromagnetic waves in different contexts, such as optics and antenna theory	Applying	Procedural Knowledge	Laboratory experiments, projects
CO6	Design and analyze complex electromagnetic systems and devices using advanced electrodynamics principles	Creating	Procedural Knowledge	Research papers, presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	ELECTRIC AND MAGNETIC FIELDS IN MATTER		12	16
	1	Polarization	2	
	2	The Field of a Polarised Object	2	
	3	The Electric Displacement; Boundary Conditions; Susceptibility, Permittivity, Dielectric Constant of Linear Dielectrics	3	
	4	Magnetisation	2	
	5	The Field of a Magnetised Object, Physical Interpretation of Bound Currents (Physical concept only), Ampère's Law in Magnetized Materials, Magnetic susceptibility and Permeability; Ferromagnetism	3	
	Sections 4.1.1 – 4.1.4, 4.2.1, 4.2.2, 4.3.1, 4.3.3, 4.4.1 of chapter 4; sections 6.1.1 – 6.1.4, 6.2.1 – 6.2.3, 6.3.1, 6.4.1, 6.4.2 of chapter 6 of Book 1			
II	ELECTRODYNAMICS		12	18
	6	Ohm's Law; Electromotive Force; Motional emf	3	
	7	Electromagnetic Induction: Faraday's Law; The Induced Electric Field; Inductance; Energy in Magnetic Fields	4	
	8	Maxwell's Equations: Electrodynamics before Maxwell; How Maxwell Fixed Ampere's Law; Maxwell's Equations; Maxwell's Equations in Matter; Boundary Conditions	5	
	Sections 7.1.1 – 7.1.3, 7.2.1 – 7.2.4, 7.3.1 – 7.3.3, 7.3.5 – 7.3.6 of chapter 7 of Book 1			

III	ELECTROMAGNETIC WAVES		8	18
	9	Waves in One Dimension, Sinusoidal waves, Polarization of Waves	2	
	10	The Wave Equations for E and B	1	
	11	Monochromatic Plane Waves	1	
	12	Poynting's Theorem	2	
	13	Energy and Momentum in Electromagnetic Waves	1	
	14	Propagation of Waves in Linear Media	1	
	Sections 9.1.1, 9.1.2, 9.1.4, 9.2.1, 9.2.2, 8.1.2, 9.2.3, 9.3.1 of Book 1			
IV	TRANSIENT CIRCUITS AND ALTERNATING CURRENTS		13	18
	15	Growth of Current in Series L-R, C-R, and L-C Circuits (Relevant portions)	2	
	16	Decay of Current in L-R, C-R and L-C Circuits (Relevant portions)	2	
	17	Alternating Current: EMF in a Coil Rotating in a Magnetic Field	2	
	18	AC Circuit Containing: R only, Inductance only, Capacitance only	2	
	19	Use of j Operator in Study of A.C. Circuits (Relevant concepts)	1	
	20	AC Circuit Containing: L and R, C and R, Parallel L and C	2	
	21	Series LCR Circuit	1	
	22	Power in AC	1	
		Sections 12.1, 12.3, 12.5, 13.1, 13.2, 13.3, 13.5 of Book 2		
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list (one from experiments 1-4 and 5 from 5-14) and 1 additional experiment, decided by the			

	teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.		
1	Verification of Faraday's law and Lenz's law of electromagnetic induction • Verify Faraday's law and Lenz's law by measuring the induced voltage across a coil subjected to the varying magnetic field. (section 7.2.1 of Book 1) • Galvanometer/ExpEYES can be used to measure the induced emf. • In the third experiment, for better coupling between the coils, use a high permeability material like iron or ferrite core, and observe the change in the induced emf. • https://expeyes.in/experiments/school-level/mutual-induction.html • Simulation: https://phet.colorado.edu/sims/html/faradays-law/latest/faradays-law_all.html		
2	Analysis of induced emf developed in a coil as a magnet dropping through it • Drop a neodymium magnet through a coil, guided through a vertical tube. • Repeat the experiment by dropping the magnet, through different heights from the coil and by changing the approaching pole. • Capture the induced emf as a function of time using ExpEYES, note the maximum value of the emf and verify Faraday's law and Lenz's law of induced emf and flux change. • Example 7.6 of Book 1 • https://expeyes.in/experiments/school-level/em-induction.html		
3	AC three phase generator • Rotate a neodymium magnet about an axis perpendicular to its dipole axis and fix three coils displaced equally from each other, i.e., 120° separated. • Analyze the induced emf developed in the coils using CRO/ExpEYES and the phase relationship between the three induced voltages. • Optional: Realize star connection (three phase four wire system) and verify the p.d. between the wires. section 13.10 of Book 2 • https://expeyes.in/experiments/school-level/ac-generator.html		
4	Demonstration of Eddy currents • Mount aluminum/copper disk as a pendulum on a horizontal axis and observe the 'viscous drag' as it swings down and		

	passes between the poles of a magnet (Can be realized using two pieces of neodymium magnet. The demonstration illustrated in Fig. 7.16 of Book 1). • https://www.youtube.com/watch?v=qTkOpprVITM OR • Form a simple pendulum with a neodymium magnet and observe the ‘viscous drag’ as it swings down when an aluminium/copper sheet/block is placed under the pendulum. • https://www.youtube.com/watch?v=VK40utGgiol • https://www.youtube.com/watch?v=SF4xjO2RN1w OR • Drop a neodymium magnet through an aluminium/copper tube and observe the delay in the fall of the magnet. Tubes of different gauge may be used for the demonstration. • Keep the two probes at diametrically opposite points of the pipe and note the emf and current when a magnet is allowed to fall through the pipe. • https://www.youtube.com/watch?v=H31K9qcmeMU		
5	Ballistic constant of the galvanometer using Hibbert’s Magnetic Standard (HMS) • Give the induced current from HMS to the BG through a series resistance. Read the deflection corresponding to the resistance in the box and hence determine the ballistic constant. • Charge a standard capacitor to its maximum capacity, with a small known voltage (using potential divider). Allow it to discharge through the BG. From the deflection in the BG, determine the charge in the capacitor and verify the relation $Q = CV$.		
6	BG-Determination of high resistance by leakage method • Charge the capacitor to its maximum capacity using a small known voltage and measure the charge stored q_0 using the BG. • Charge it again and allow it to discharge through a high resistance, R for a small interval of time, T. After this the remaining charge, q is measured using the BG. • Using the values of q_0, q, T and C, calculate the value of R.		
7	Mutual inductance and coefficient of coupling using Anderson’s bridge • Connect the two coils of known self-inductances, L_1 and L_2 in series along with the resistance box in one of the arms of the Anderson’s bridge. Keep one coil flat over the other so that the configuration gives maximum mutual inductance between the two. • Determine the self-inductance $L' = L_1 + L_2 + 2M$ of the series combination using the null method. Reverse the coupling		

	between the coils by reversing the connections in one of the coils and once again determine the self-inductance $L''=L_1+L_2-2M$ of the combination. • Compute $M=(L'-L'')/4$ and $k = M/\sqrt{L_1L_2}$		
8	Parallel plate capacitor. (a) verify the relationship between capacitance and the area of the plates (b) determination of dielectric constant of thin dielectric sheet • Form a parallel plate capacitor with dielectric material filled between the plates. • Multimeter/ ExpEYES can be used to measure the capacitance. (For a significantly measurable value of the capacitance, use plates of dimension 10cmx10cm, or greater) • Change the area of the capacitor plates and verify the relationship of the capacitance on the area (Using the same set of plates, the area can be changed by varying the overlapping region of the plates) • By measuring the capacitance for different areas of the capacitor plates and (or) thickness of the dielectric material, determine the dielectric constant of the given material/liquid. • https://www.youtube.com/watch?v=IKfIkUuFT-U		
9	Brewster's law experiment, determination of angle of polarisation and refractive index • Experimental arrangement- Sodium vapour lamp, Spectrometer, Polarizer (Graduated on 360° rotating) coupled in front of the spectrometer telescope, prism or glass plate. • Get the angle of incidence corresponding to the minimum intensity of light and hence calculate the refractive index of the material. • https://www.youtube.com/watch?v=f2A8sM1xhbQ		
10	RC and RL transients - determination of capacitance and inductance • Apply a voltage step to a series RC/RL circuit and record the resulting voltage variation across the capacitor/inductor. • Get the value of time constant by an exponential fit to the data. • Repeat the experiment for different resistances. • https://expeyes.in/experiments/electrical/rctransient.html • https://expeyes.in/experiments/electrical/rltransient.html		
11	RL and RC series AC circuits- Phase relationships of voltage across the elements		

		- Using a CRO/ ExpEYES, verify the phase relationship between voltage across the inductor/capacitor and the current. - Note the phase difference between the applied voltage and current and determine the value of inductance/capacitance. OR - Note the peak voltage and current and determine the value of inductance/capacitance. https://expeyes.in/experiments/electrical/rcsteady.html https://expeyes.in/experiments/electrical/rlsteady.html https://expeyes.in/experiments/school-level/ac-rc.html https://expeyes.in/experiments/school-level/ac-rl.html		
12	Series LCR circuits-Determination of resonance frequency, quality factor and bandwidth	- The frequency of the signal generator is changed in steps and the corresponding voltage across the resistance is noted. - From the graph drawn for current against frequency, find the frequency corresponding to maximum voltage- resonant frequency. Also find the bandwidth and quality factor - CRO/Multimeter/ExpEYES can be used. https://expeyes.in/experiments/electrical/rlcsteady.html		
13	Simulation of the behavior of RC and RL circuits under AC and DC	- Simulate the behavior of RC and RL circuits under AC and DC sources. - Section 8.3, 8.4 & 9.3 of Book 3 can be referred.		
14	Simulation of the behavior of RC and RL circuits under AC and DC	- Simulate the behavior of RC and RL circuits under AC and DC sources. - Section 9.3 of Book 3 can be referred to.		

Book for Reference:

1. Introduction to Electrodynamics (5th Edn.) by David J Griffiths, Cambridge University Press (Book 1)
2. Electricity and Magnetism (10 Edn.) by R Murugesan, S. Chand and Company (Book 2)
3. Electrodynamics Tutorials with Python Simulations by Taejoon Kouh, Minjoon Kouh -CRC Press 1st Edition (Book 3)
4. Electricity and Magnetism, Berkeley Physics Course Vol.2, by E M Purcell, McGraw Hill Edn.
5. Electricity and Magnetism, by D C Pandey, Arihand Prakashan Series
6. Electrodynamics Made Simple - e book by E D Dias and Santhosh P Jose
<https://store.pothi.com/book/ebook-dr-dias-e-d-electrodynamics-made-simple/>

7. Classical Electromagnetism by H C Verma, Bharathi Bhavan Publishers and Distributors
8. The Feynman Lectures on Physics, Vol - 2, Pearson Education India

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO 4	PO 5	PO 6	PO 7
CO 1	2	1	2	0	2	0	2	1	1	0	2	1	2
CO 2	0	2	3	0	2	0	1	3	2	0	2	2	2
CO 3	2	0	3	0	2	1	2	2	2	0	3	2	3
CO 4	2	2	2	1	3	1	2	2	2	1	3	2	3
CO 5	1	3	2	1	3	2	1	3	2	2	3	3	3
CO 6	3	2	3	2	3	2	2	3	3	3	3	3	3

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	